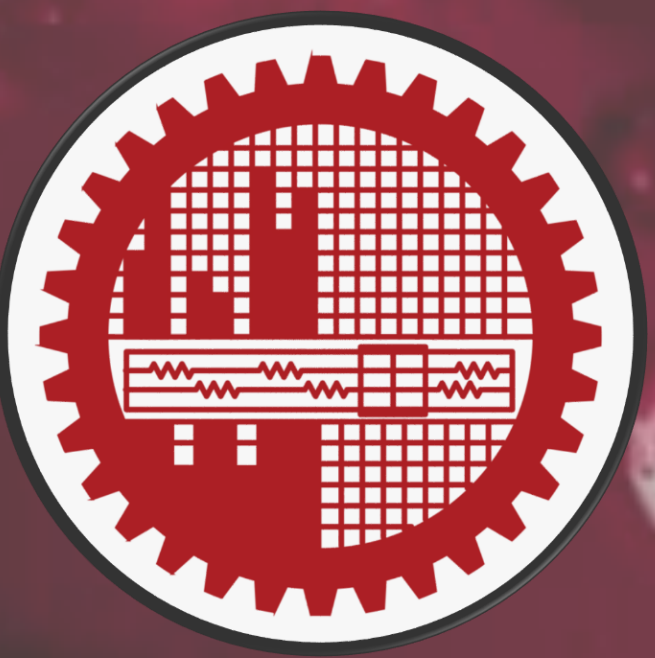


A Low Cost Automated Dengue Fluid Management System



Auto-Drip

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Abstract

Dengue fever remains a significant public health concern especially in this subtropical region, and the need for effective and cost-efficient solutions to manage its complications is increasingly pressing. In 2023, Bangladesh witnessed its highest number of dengue-related deaths in 23 years, with 321,179 hospitalizations and 1,705 fatalities. As of June 28, 2025, there have been 9,484 dengue cases and 41 deaths reported this year.

Although effective fluid management is critical for treating dengue patients, frequent monitoring of IV flow rates is challenging due to limited healthcare personnel. To address this issue, we have developed AutoDrip, an advanced Dengue Fluid Management System designed to automate and optimize IV fluid delivery. The system automates IV fluid management by:

- **Detecting** the current flow rate using an IR sensor.
- **Calculating** the required flow rate based on patient vitals or user input.
- **Adjusting** the flow rate automatically via a stepper motor.

The system features two modes: normal mode and dengue mode.

Problem Statement

Dengue fever, caused by the dengue virus, leads to complications like **plasma leakage**, which can result in shock and organ failure if not managed properly. Effective **fluid management** is critical for Type B dengue patients to maintain blood volume.



Bangladesh suffers from 76 percent shortage of nurses

The country needs more than 3 lakh nurses. But the number of registered nurses here was 73,043 till April this year

Fig 1 : THE BUSINESS STANDARD,Kamran Siddiqui, 02-05-2020

However, Bangladesh faces severe healthcare manpower shortages, with a **population-to-nurse ratio of 5000:1** and a **bed-to-nurse ratio of 13:1**, making timely treatment difficult.

Methodology

Device Description

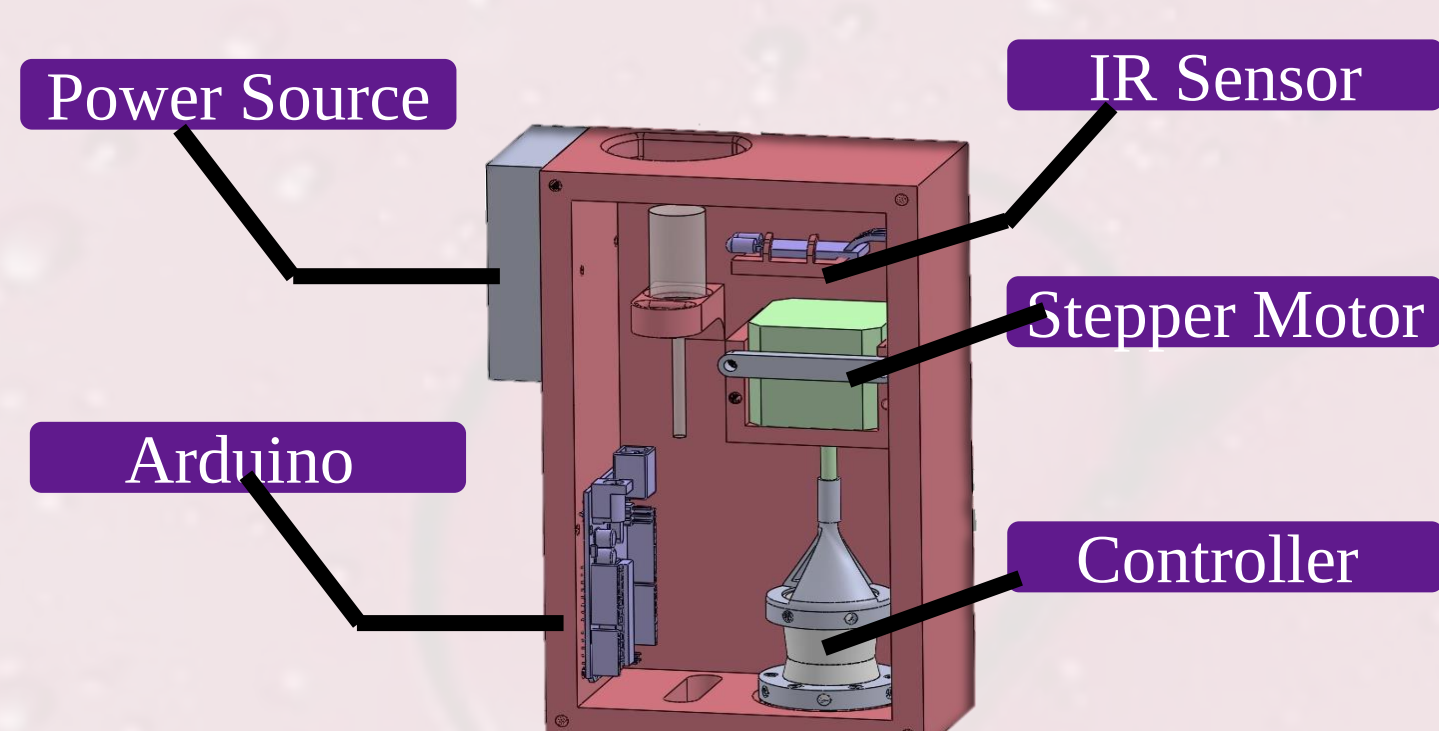


Fig 2 : Device Morphology

1) Desired Flow Rate Determination

- ❑ **Normal Mode:** Allows clinicians to manually input a specific flow rate using an integrated keypad.
- ❑ **Dengue Mode:** Automatically calculates the optimal flow rate based on patient vitals (e.g., weight, patient type) to comply with clinical guidelines.

2) Flowrate Detection:

A non-contact Infrared (IR) Sensor is positioned on the IV drip chamber to provide real-time measurement of the current drop rate (drops/minute).

Fig 3 :IR Sensor

Methodology

3) Automated Control and Feedback Loop:

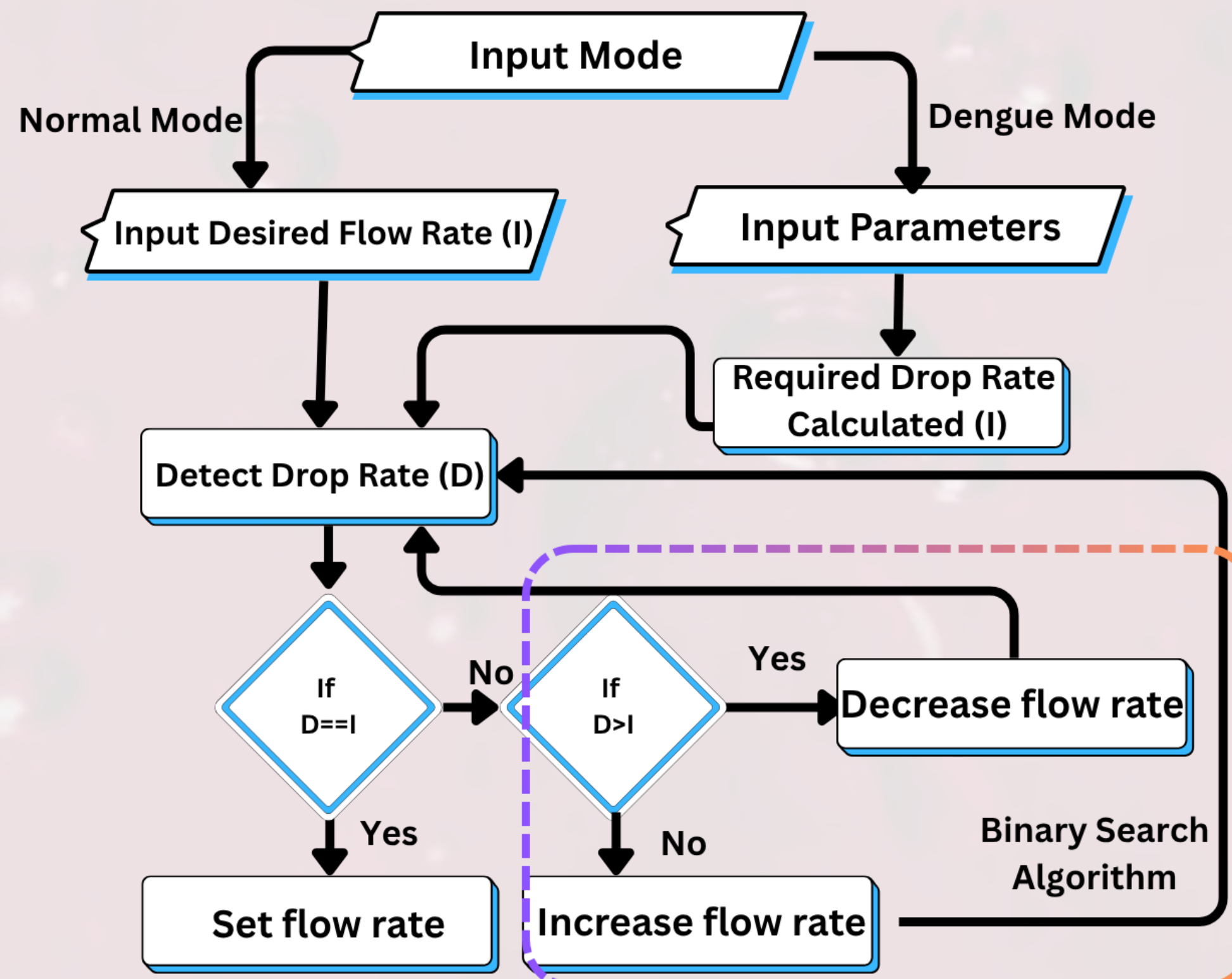


Fig 4 : Flowrate of the working principle

- The system's core logic compares the real-time rate from the IR sensor to the desired target rate.
- A Binary Search Algorithm is implemented to efficiently find the correct rotational position for the stepper motor, allowing the system to rapidly converge on the target flow rate (typically within 8 iterations).

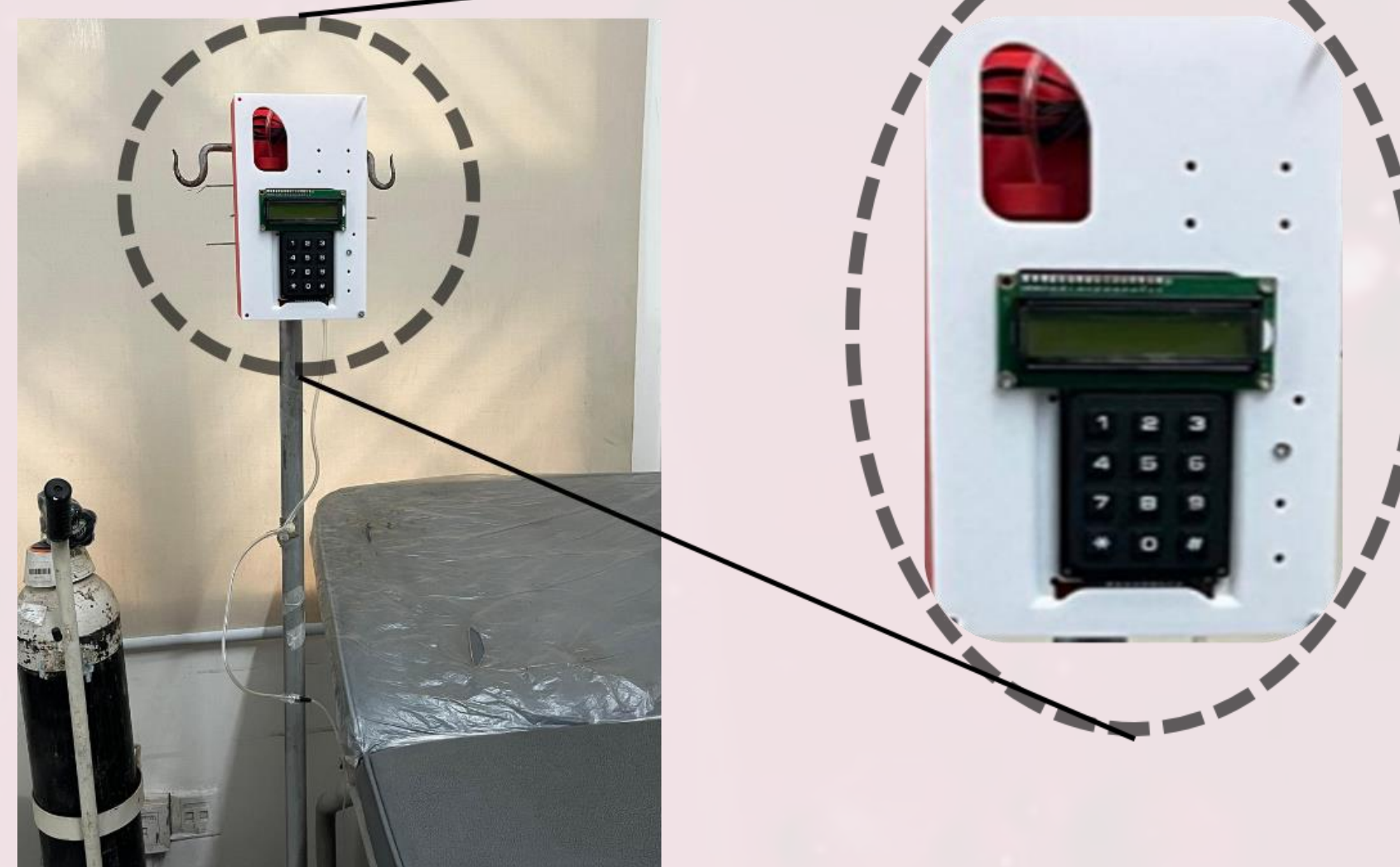


Fig 5 : Final Device

4) Flowrate Control:

We rejected linear constriction methods (like rack and pinion) due to torque and stability issues. The final design uses a Stepper Motor to precisely rotate a standard, off-the-shelf Dial Flow Controller. A custom 3D-printed Conical Coupler was designed to ensure a strong, perfectly aligned connection between the motor and the dial.



Fig 6 : Flowrate Controller

5) Fail-Safe System:

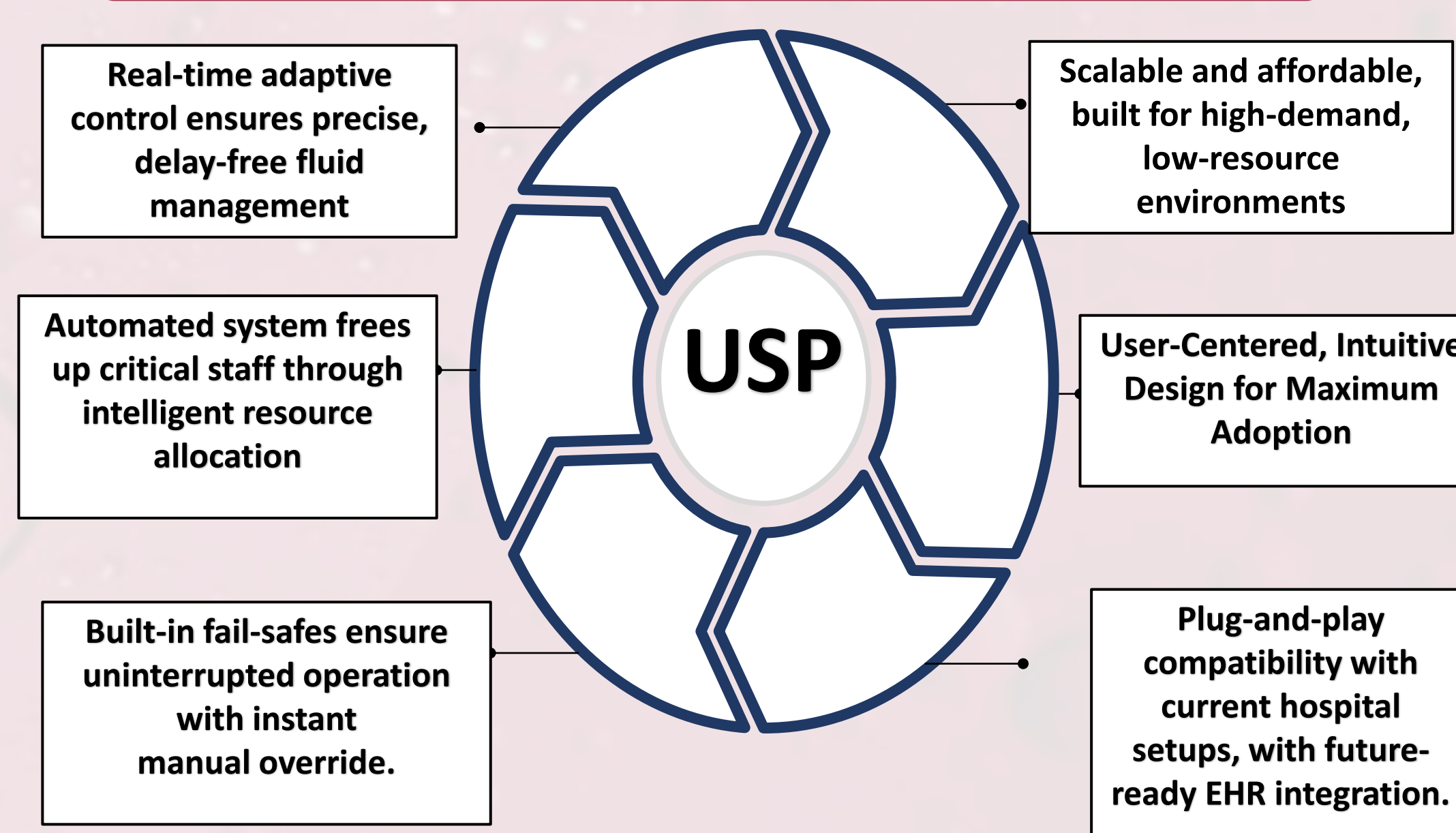
➤ Device Fall Detection:

If the device accidentally falls, an Accelerometer detects the movement and sends an immediate notification, ensuring prompt corrective action.

➤ Saline Bottle Monitoring:

The system monitors the saline bottle and alerts the user when it is empty. This prevents interruptions in fluid delivery, especially during critical treatments..

Unique Selling Point



Result & Analysis

IR Sensor Validation: The graph shows the IR sensor's drop detection with minimal error, confirming its accuracy in comparing actual and expected rates, forming the basis of the feedback system.

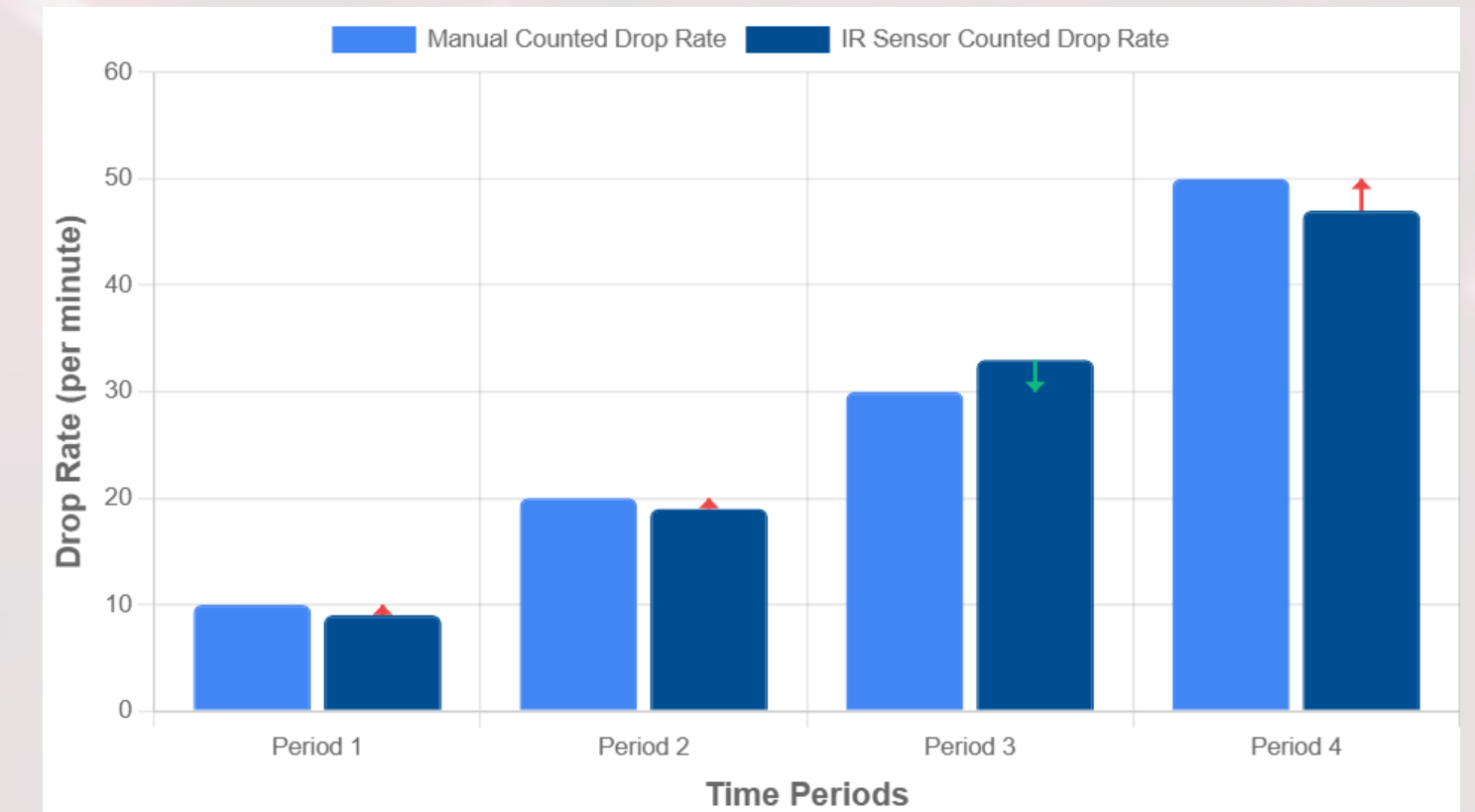


Fig 7 : Drop Rate manual vs IR sensor

Binary Search Algorithm Validation: The graph illustrates that the algorithm reaches the target flow rate within 8 iterations for all test cases, highlighting its efficiency.

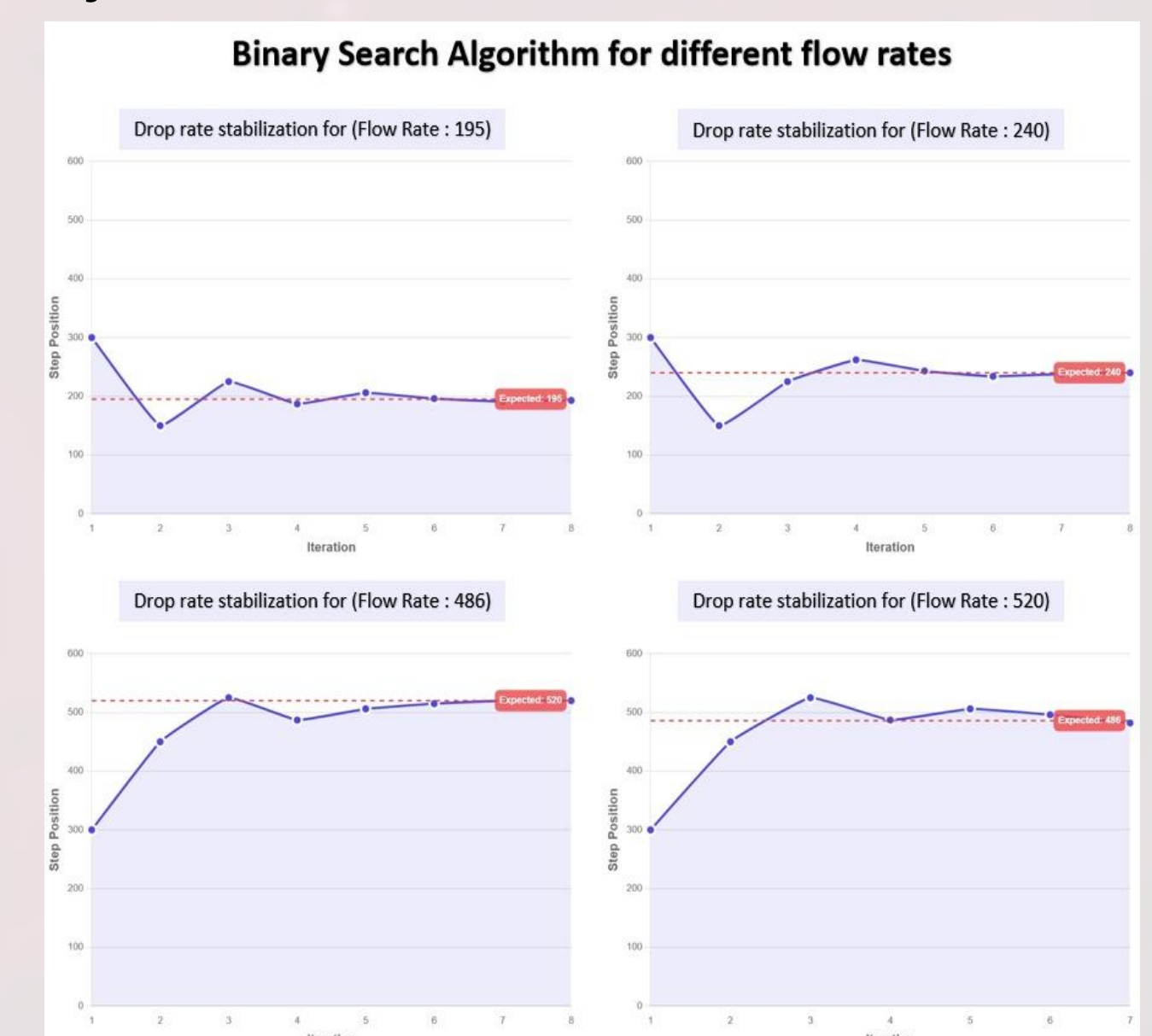


Fig 8 : Validation of Binary Search Algorithm

Flow Rate Calculation: The graph shows that the device's 48-hour flow rate matches the output of the DGHS-approved Dengue Drops App, validating its reliability.

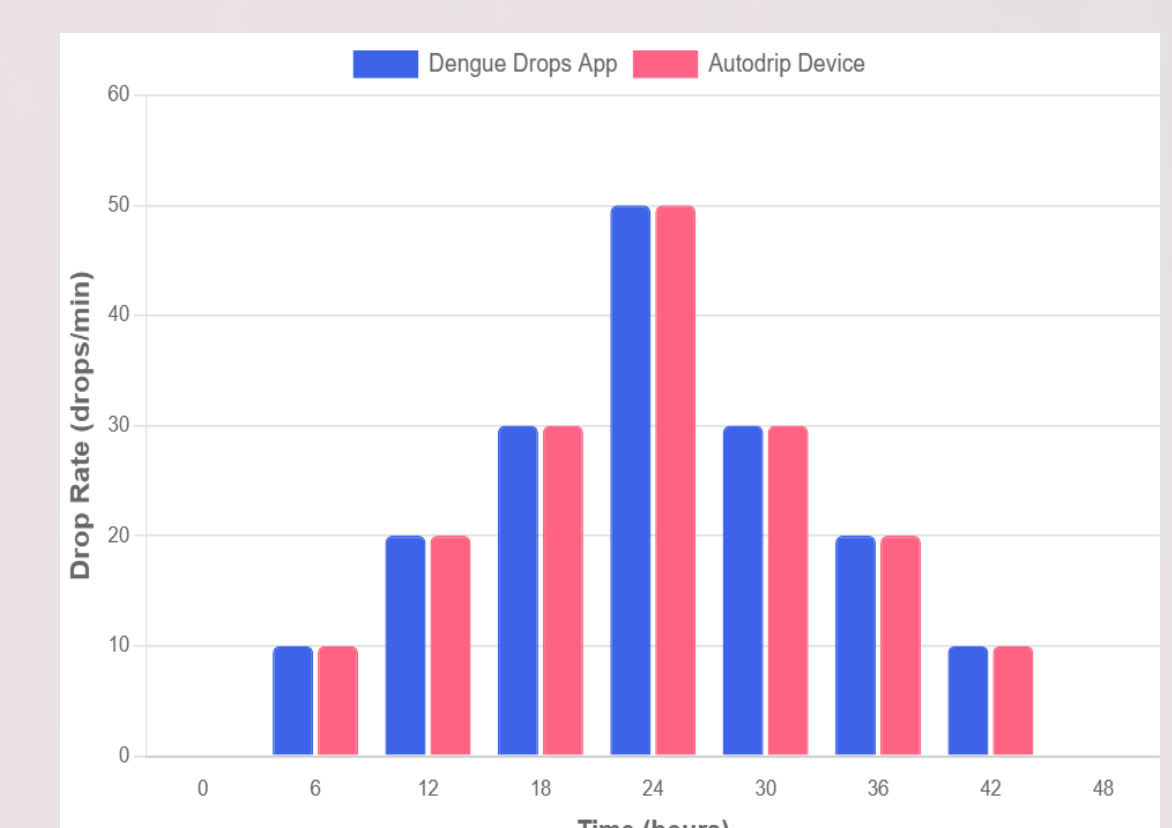


Fig 9: Data validation by Dengue Drops app

Discussion

The device has the potential to revolutionize the automation of our healthcare system. It offers a **simple yet highly efficient solution** to address challenges during pandemic situations. Equipped with a **responsive feedback mechanism**, the device can seamlessly adapt to any abrupt changes in flow rate, ensuring consistent and accurate fluid delivery. Furthermore, its **embedded alert system** proactively notifies users of any issues, guaranteeing safety and reliability. With these features, the device stands out as a **safe, effective, and indispensable tool** for modern healthcare settings.

Reference & Acknowledgement

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